Introduction

A contingency table is a cross-tabulation treated as the input to a statistical test of independence. The counts in the cells, the expected counts under the independence null, and the standardised residuals that localise any deviation are its three essential components.

Prerequisites

The reader should know what a cross-tabulation is.

Theory

For an $r \times c$ contingency table with observed counts n_{ij}, row totals n_{i+}, column totals n_{+j}, and grand total n:

- **Expected count** under independence: $E_{ij} = n_{i+} \cdot n_{+j}/n$.
- **Pearson residual**: $e_{ij} = (n_{ij} - E_{ij})/\sqrt{E_{ij}}$.
- **Standardised residual**: Pearson residual divided by $\sqrt{(1 - n_{i+}/n)(1 - n_{+j}/n)}$; approximately standard normal under the null.
- **Chi-squared statistic**: $\chi^2 = \sum(n_{ij} - E_{ij})^2/E_{ij}$; under the null, it follows χ^2 with $(r-1)(c-1)$ df.

Standardised residuals beyond ± 2 flag cells that substantially exceed or fall short of independence.

Assumptions

For the chi-squared approximation: independent observations and all expected counts ≥ 5. For small expected counts, use Fisher's exact test.

R Implementation

```r
library(vcd)

cohort <- matrix(c(24, 18, 36, 32, 18, 42, 16, 24, 40),
                 nrow = 3, byrow = TRUE,
                 dimnames = list(Stage = c("I", "II", "III"),
                                 Response = c("CR", "PR", "SD")))
cohort

chi <- chisq.test(cohort)
chi
chi$expected
chi$residuals          # Pearson residuals
chi$stdres             # standardised residuals

# Visual representation of residuals
mosaic(cohort, shade = TRUE, legend = TRUE)
```

Output & Results

```
        Response
Stage    CR PR SD
  I      24 18 36
  II     32 18 42
  III    16 24 40
```

```
Expected:
       CR   PR   SD
  I   23.5 18.3 36.2
  II  27.4 21.4 43.2
  III 21.1 16.5 33.4

Standardised residuals:
       CR    PR    SD
  I   0.15 -0.09 -0.08
  II  1.32 -1.02 -0.30
  III -1.50  2.18  1.53
```

No cell's standardised residual exceeds ±2.5, so none drives a substantial departure from independence. The chi-squared p-value ($p \approx 0.22$ here) is non-significant.

Interpretation

Report the contingency table alongside the chi-squared statistic and a measure of association (Cramer's V). Standardised residuals are reported when the omnibus test is significant and the investigator wants to localise the effect.

Practical Tips

- `chisq.test()` returns the expected counts, Pearson residuals, and standardised residuals as attributes.
- Mosaic plots (`vcd::mosaic`) encode cell deviations with colour – a visual alternative to residual tables.
- For small tables with sparse cells, `fisher.test()` (or Monte Carlo chi-squared) is required.
- For stratified 2x2 tables across a third variable, use Cochran-Mantel-Haenszel (`mantelhaen.test`).
- `vcd::assocstats()` returns phi, contingency coefficient, and Cramer's V in one call.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1